Korea Technology and Information Promotion Agency for SMEs

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

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2. Technology Overview

Title of Proposed Technology	Tailored beam shapes for Through Glass Via laser processing with real time, non-destructive inspection and subsequent, in-line repair process			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display (√), Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio ( ), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI ( ), Next-gen Communication ( ), Advanced Robotics & Manufacturing ( ), Quantum ( )			
Industrial Technology Classification [1)]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Mechanical Materials		Electrical Electronics	
	Medium Category	Small Category	Medium Category	Small Category
	Precision manufacturing machinery	Optical Energy–Based Processing Machines	Optical Application Devices	Laser Processing Machines
Concept of Proposed Technology	**Core concept of the proposed technology:** • Non-destructive inspection technology of in-volume USP laser modifications in glass for determination of etching behavior and resulting hole shape, surface roughness, taper and quality for Through Glass Via (TGV) processing to bridge the ever-widening gap between on-chip (front-end, FE) and back-end (BE) of modern chip technology • Result: 100% of the TGVs on a glass substrate are within the specified tolerances **Main Functions:** • Vision inspection solution before and after etching with tailored optics • Automated recognition and marking of blind modifications • Tailored temporal and spatial beam shapes for adapting TGV processing and inspection results, as well as parameters for in-line repair processes **Potential uses of proposed technology:** • Development of advanced packaging solution with Through Glass Via (TGV) to bridge the			

Rationale for Joint R&D Project	ever-widening gap between on-chip (front-end, FE) and back-end (BE) of modern chip technology • Contacting of quantum components like ion traps for quantum computers
	The necessity for this joint R&D project arises from the innovative potential of small and medium-sized enterprises (SMEs) in Korea, particularly in the realm of in situ analysis for Through Glass Via (TGV) processing. These companies are currently undertaking numerous efforts to implement non-destructive quality control methods. However, the TGV processing domain is predominantly advanced by larger companies, resulting in a significant knowledge gap regarding TGV processing technologies among SMEs, which are primarily focused on in situ analysis of modifications. To remain competitive in the market, these SMEs must be equipped to offer comprehensive machine technologies for TGV analysis, including subsequent repair processes. This capability is essential for their growth in an increasingly competitive environment. Collaboration with Fraunhofer ILT is crucial in this context. The Fraunhofer ILT possesses extensive expertise in ultrashort pulsed (USP) laser in-volume structuring, which is vital for the effective processing of TGVs. Additionally, the Institute's knowledge in USP laser material processing of glass and its ongoing development of flexible tailored spatial and temporal beam shaping is invaluable. Thus, a partnership between Korean SMEs and the Fraunhofer ILT is essential to ensure the development of independent products and to secure a sustainable market presence of the SMEs. By leveraging the advanced technologies and knowledge of the Fraunhofer ILT, these companies can enhance their capabilities and contribute significantly to the advancement of TGV processing technologies.
Key Words	1. Through Glass Via laser processing 2. Real-time, non-destructive, optical inspection technology 3. Tailored temporal and spatial beam shaping 4. In-line laser repair process

3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	Start of the Project	Level (≤4) depending on SME technology	End of the Project	Level (6-7) depending on SME technology
R&D Objectives	• In situ, non-destructive, optical inspection technology of laser modifications and Through Holes leading to determination of blind holes • TGV processing and laser-based repair process after inspection with flexible temporal and spatial beam shaping on panel sizes up to 515 mm x 510 mm x 1.0 mm • Machine integration of above aspects including optical and mechanical design			

R&D Target Specifications	• Large scale non-destructive, optical inspection at same speed as laser writing speed (>5000 holes/sec)
	• Automated recognition of insufficient modifications, prediction of etching behavior and determination of blind holes
	• Data preparation (position, classification of modification type, necessary laser parameters for repair process, etc.) to guide subsequent laser repair process
	• Laser-repair process for blind holes (before and after etching):
	  ○ Glass thickness: 0.5 mm ~1.0 mm
	  ○ Hole diameter: 10 µm ~100 µm
	  ○ Aspect ratio > 1: 20 ~ > 1:50
	  ○ Crack-free
	  ○ Surface roughness Ra<200 nm			

R&D Plan	**Key tasks:**
	• On-going exchange of results, samples and knowledge gained (SME & ILT task)
	• Development of non-destructive, optical inspection technology of TGVs (SME task)
	• Classification of different modification types and occurring insufficient in-volume modifications (SME task); Provision of samples with various insufficient modifications, including the most common errors (ILT task)
	• Process development of laser repair process based on the recognized insufficient modification type (ILT task)
	• Machine integration and testing at SME in Korea (SME & ILT task)
	Methodologies:
	• Methodologies of non-destructive, optical inspection technology depends on applying SME's approach
	• Laser process development for USP laser repair process includes variation of:
	  ○ Wavelength (IR-UV)
	  ○ Pulse duration (sub 100 fs to several ps)
	  ○ Bursts (MHz/GHz)
	  ○ Flexible, spatial beam shaping with diffractive neural networks (DNN)
	  ○ Double-SLM approach for increased freedom of intensity distribution
	• In-Volume modifications as well as drilling are investigated for USP laser repair process
	Expected milestones:
	• M1 (after 12 months): Non-destructive, optical inspection technology can classify >99% of insufficient modifications and blind holes
	• M2 (after 24 months): Laser repair process of insufficient modifications and blind holes is demonstrated
	• M3 (after 36 months): Machine prototype is built and tested at SME
	Timeline:
	• Approximated project duration: 36 months
	• Year 1:
	  ○ Focus on development of non-destructive, optical inspection technology
	  ○ Preparation of laser processing and beam shaping
	• Year 2:			

	○ Focus on USP laser repair process in combination with output from inspections ○ Further automation of inspection technology and layout for prototype machine • Year 3: ○ Assembling and testing of prototype machine for inspection and repair process ○ Demonstration of automated laser repair process

4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	Role: Experts in laser-materials processing and optical technology (See referenes) • Selective Laser-induced Etching • Scaling (multibeams, elongated beams, bursts, fast beam deflection) • Quality improvement (Higher precision, lower stress and crack formation) • Integration of laser-materials processing Responsibilities: • Provision of test samples for inspection tests • Development of USP laser repair process • Tailored beam shaping with DNNs and two SLMs Capabilities: • 420 employees with more than 11,000 m^2 of usable floor space • >15 high precision USP laser machines (USP laser power up to 1 kW, wavelength from IR to DUV, pulse duration down to 50 fs, flexible beam shaping with >100 beamlets) • In-situ pump-probe microscopy systems (time resolution <300 fs with ranges from fs to ms regime) • Well-equipped in-house metrology (SEM, AFM, LSM, WIM) Contributions: • IP of SLE (from laser-materials interaction, glass type dependent process regimes, beam shaping, scan strategies and etching solutions) • High-speed, high-aspect ratio TGV processing development • Test of methods for homogeneity and etching behavior inspection before etching • Support for optical layout of inspection optics • Technology transfer of USP laser repair process to machine concept of Korean SME • Training of workforce at SME to implement and use USP laser repair process • Access to ILT's supplier and industry partner network
Expected R&R of Korean SME Partner	Role: Experts in non-destructive, optical inspection technology of glass substrate and silicon wafer • Optical system design for vision inspection of glass substrate and silicon wafer • Building integrated laser-based setups for microelectronic and display industry Contributions: • Background IP in their fields (e.g. novel optical system for TGV vision inspection) • AI-based tools for process analysis, quality investigation and inspection • Provision of relevant technical requirements The Korean SME will lead commercialization efforts encompassing regulatory compliance, IP management, and market strategies

5. Expected Outcomes/Impact

Expected Outcomes and Impact	• TGV structuring machine prototype with integrated vision inspection before etching • Inspection technology in combination with repair process lead to 100% of the TGVs on a glass substrate are within the specified tolerances → High impact on current TGV market → Commercialization of inspection machine to chip manufacturers • It can be expected that a successful project leads to an invention or patent. → Secured IP leads to strengthened position of SME in international competitive envirnment • Joint publications in industrial or open access journals as well as on international conferences and fairs • Cooperation in future projects both on a public as well as a bilateral basis

6. References

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☐ Terms:

1) Industrial Technology Classification

· Classification system according to Industrial Technology Classification Table (**Appendix 1**, based on Article 16 of the Common Operational Guidelines for the Industrial Technology Innovation Project in Korea).

2) Technology Readiness Level (TRL)

Description		
·A consistent and objective indicator of the technological maturity of key element technologies		
	Basic Research Stage (1-2)	1st Stage: Basic theories and experiments 2nd Stage: Proof of Concept, such as practical ideas or patents
	Experimental Stage (3-4)	3rd Stage: Verification of basic performance at a laboratory scale 4th Stage: Evaluation of key performance of materials, components, and systems at a laboratory scale
	Prototype Stage (5-6)	5th Stage: Production and performance evaluation of prototypes of finalized materials, components, and systems 6th Stage: Production and performance evaluation of prototypes at a pilot scale
	Productization Stage (7-8)	7th Stage: Reliability testing and evaluation by potential customers 8th Stage: Prototype certification and standardization
	Commercialization Stage (9)	9th Stage: Commercialization